

WeRead the Instructions before proceeding:

1. This is a **closed-book exam**. You can use a **calculator** if necessary.
2. **Please Write/Draw legibly!** If we can't understand what you have written, we can't grade it.
3. **Don't use Pencils** for answering/drawing. The final answer **must** be in blue or black ink.
4. Mention the question number before the answer.
5. Read the instructions carefully.

1. a) What is PEAS for grocery store scanners? **(5 Marks)**

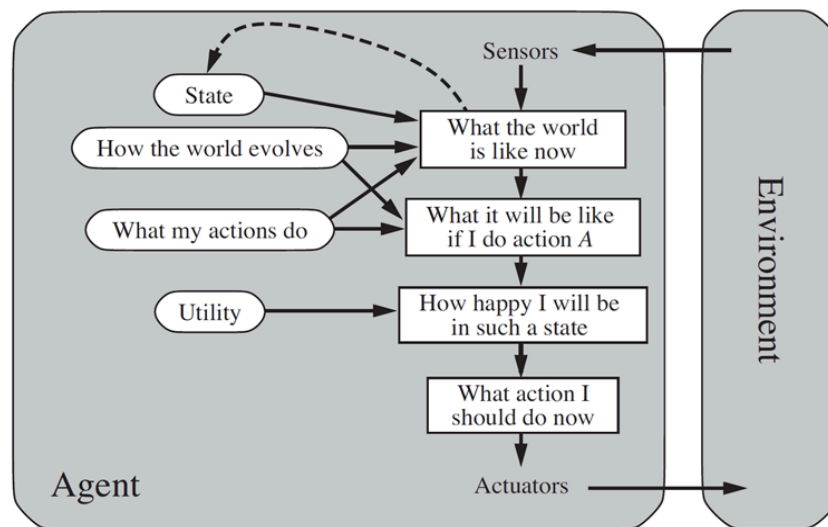
Performance: Read barcode properly, quickly and accurately collect data about products.---**2M**

Environment: Shopping area, Mall etc.---**1M**

Actuators: Screen display (Money, products information)---**1M**

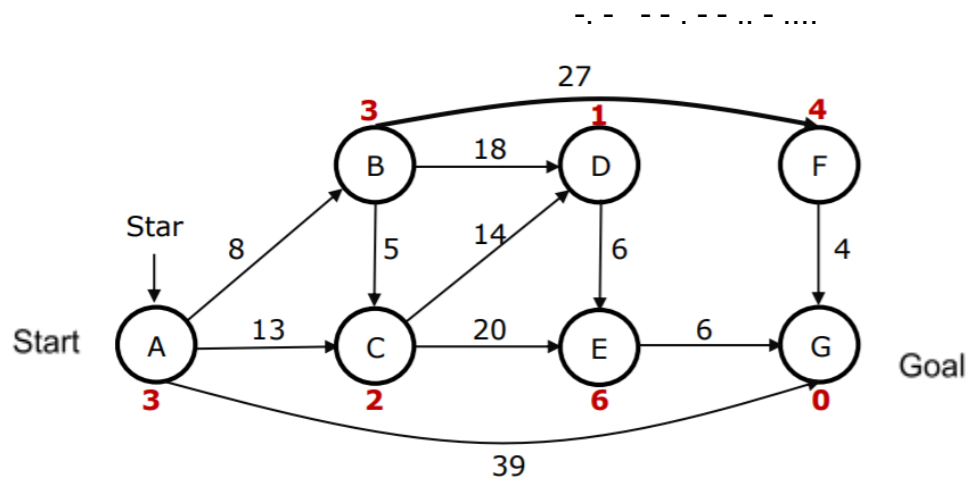
Sensors: Camera for code readers---**1M**

- b) Draw a Utility-based reflex agent block diagram and explain with an example **(5 Marks)**



The figure will get 3 Mark And example will get 2 Marks

2. Use the following tree to indicate the order in which nodes are expanded, for Greedy Best First Search, and A* search. Assume that G is the only goal node, and A is the initial node. Here, path costs are shown to the right of each path, g = cost of path so far, h = estimate of remaining cost to goal, and f = estimate of total path cost. **(10 Marks; 5 marks for each algorithm)**

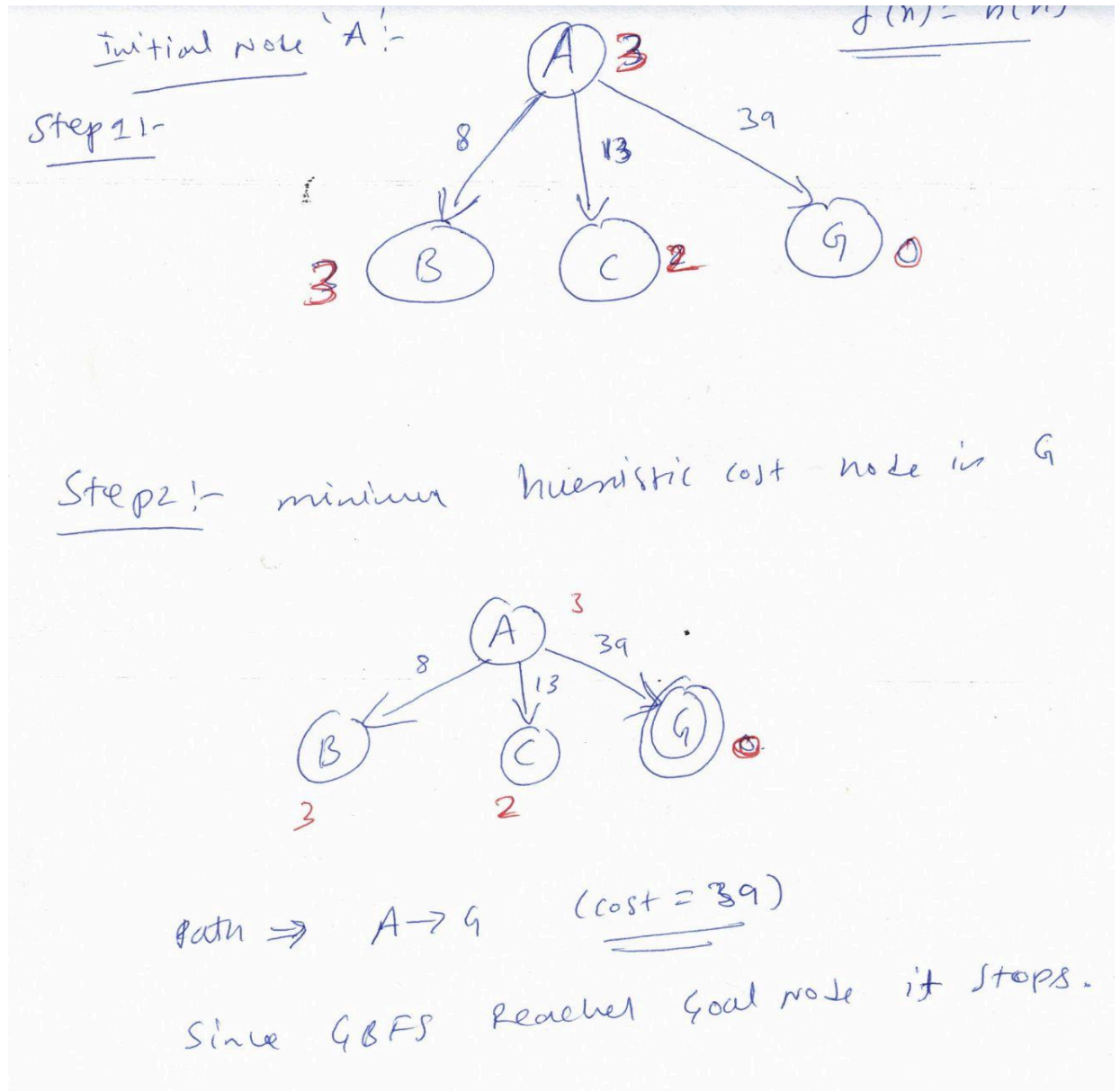


Note: Step-by-step expansion needs to be shown, any direct answers will be awarded with zero marks.

Answer:

For GBFS; Expanding A will get only 1 Mark. If you expand correctly in the next step you will get the remaining 4 Marks. Any other answers will get zeros

Greedy Best First Search:

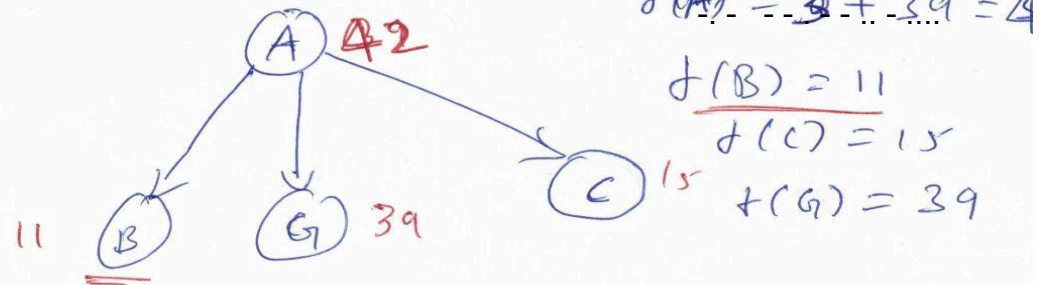


A* Star:

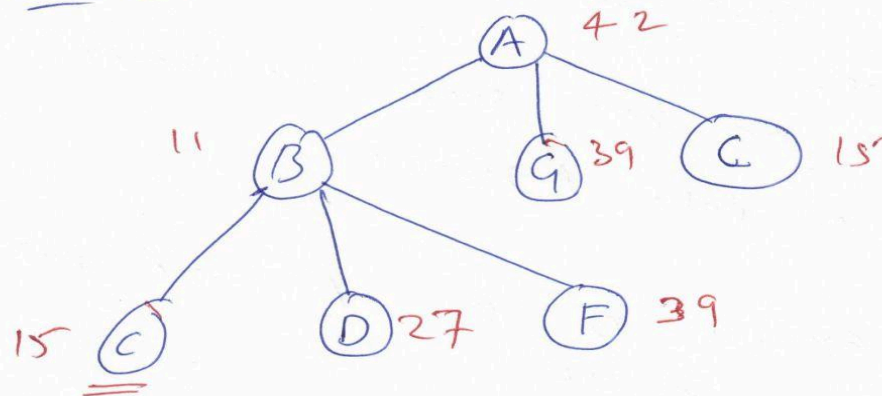
For A* Each Step of correct expansion will get 0.5 Marks. Writing the Optimal path and cost will get 1 Mark.

Any irrelevant expansions and answers will get zero.

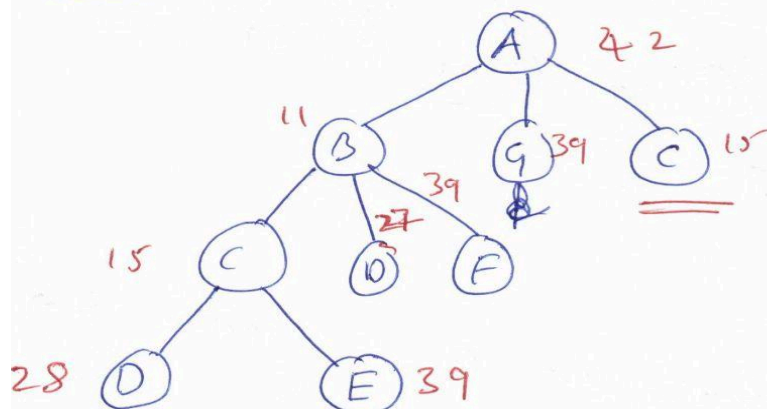
Star Step 1:- Expand 'A' $f(n) = g(n)$



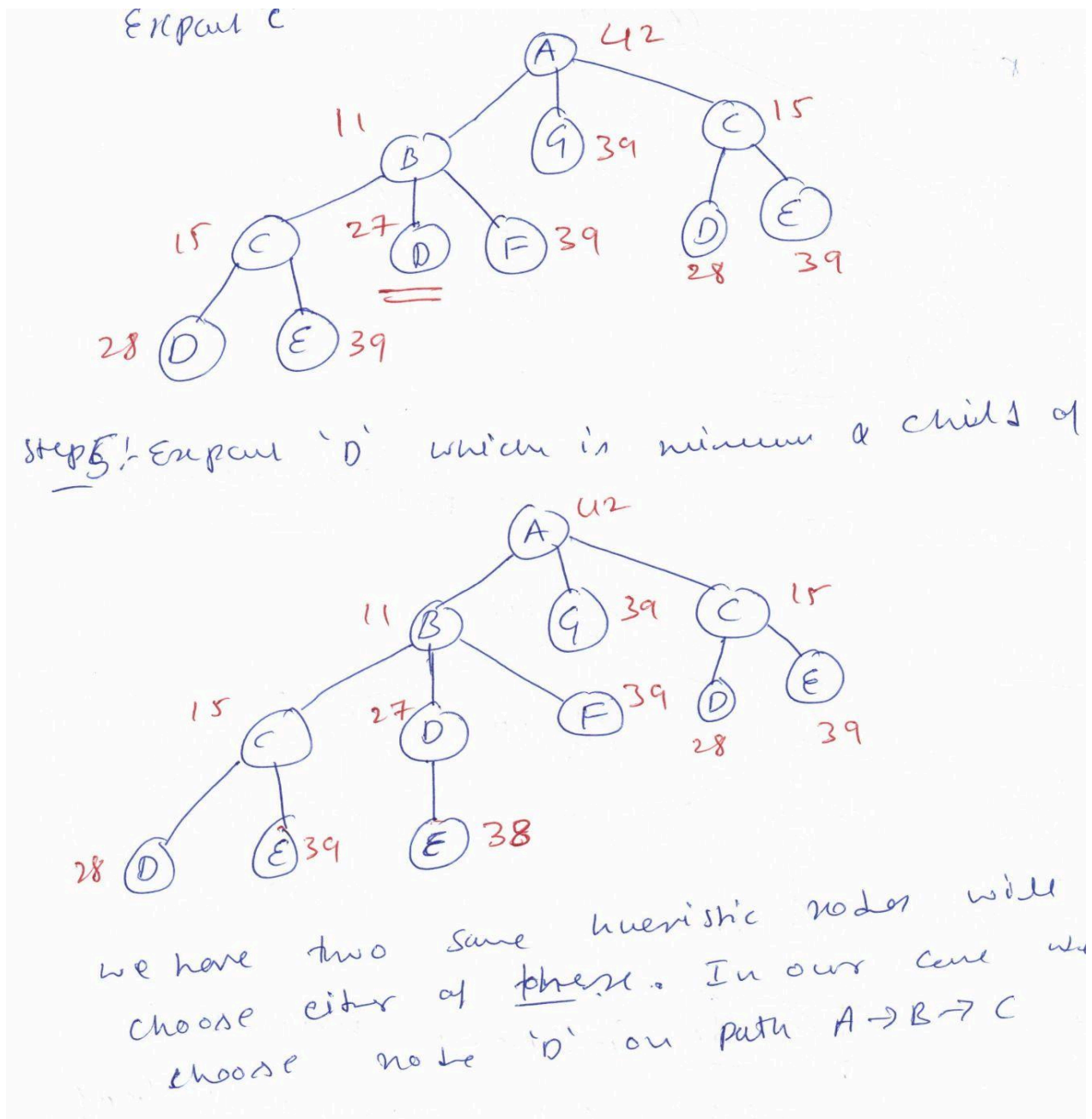
Step 2:- Expand 'B'

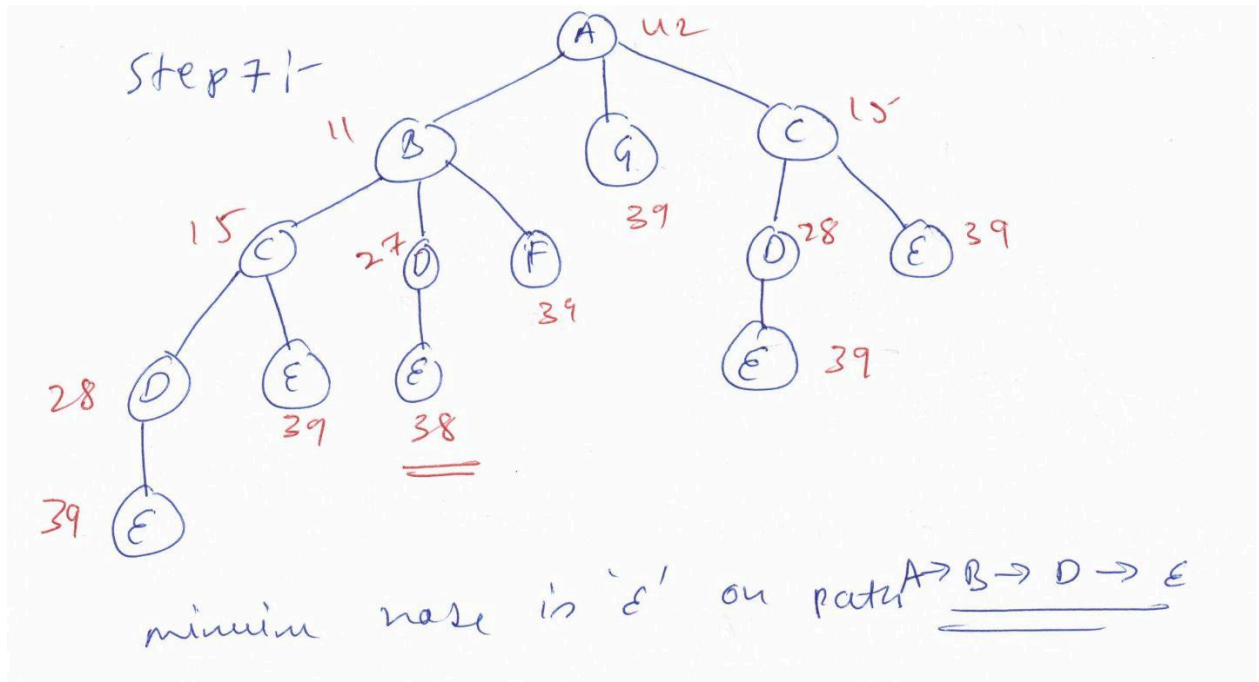
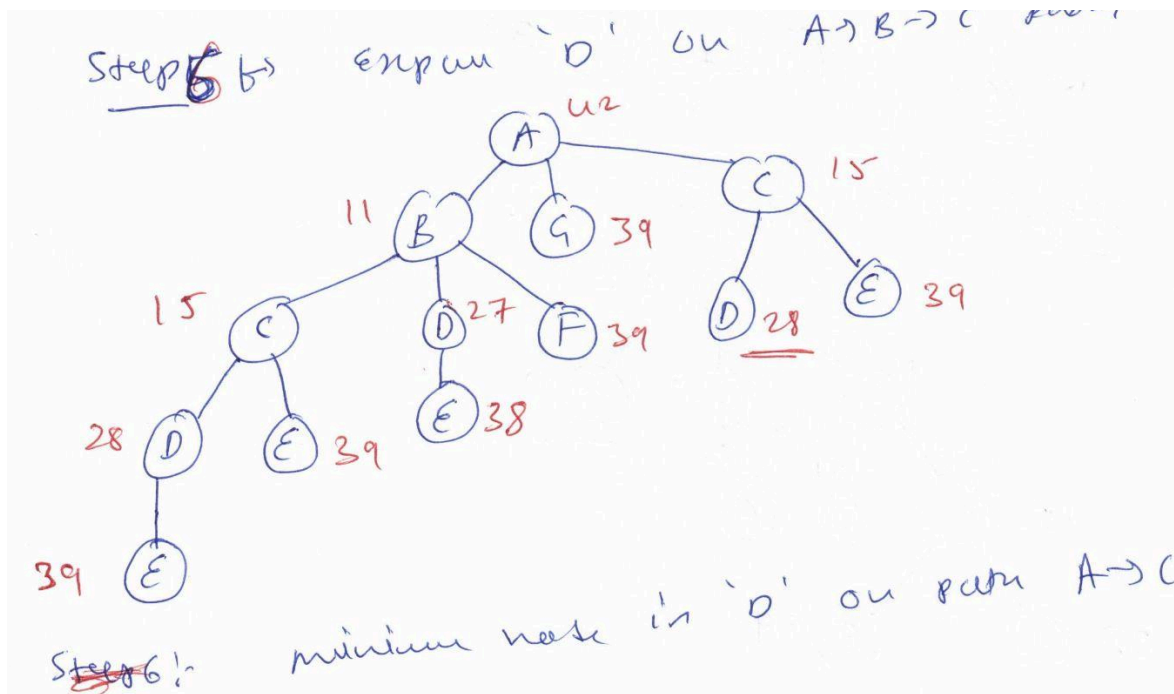


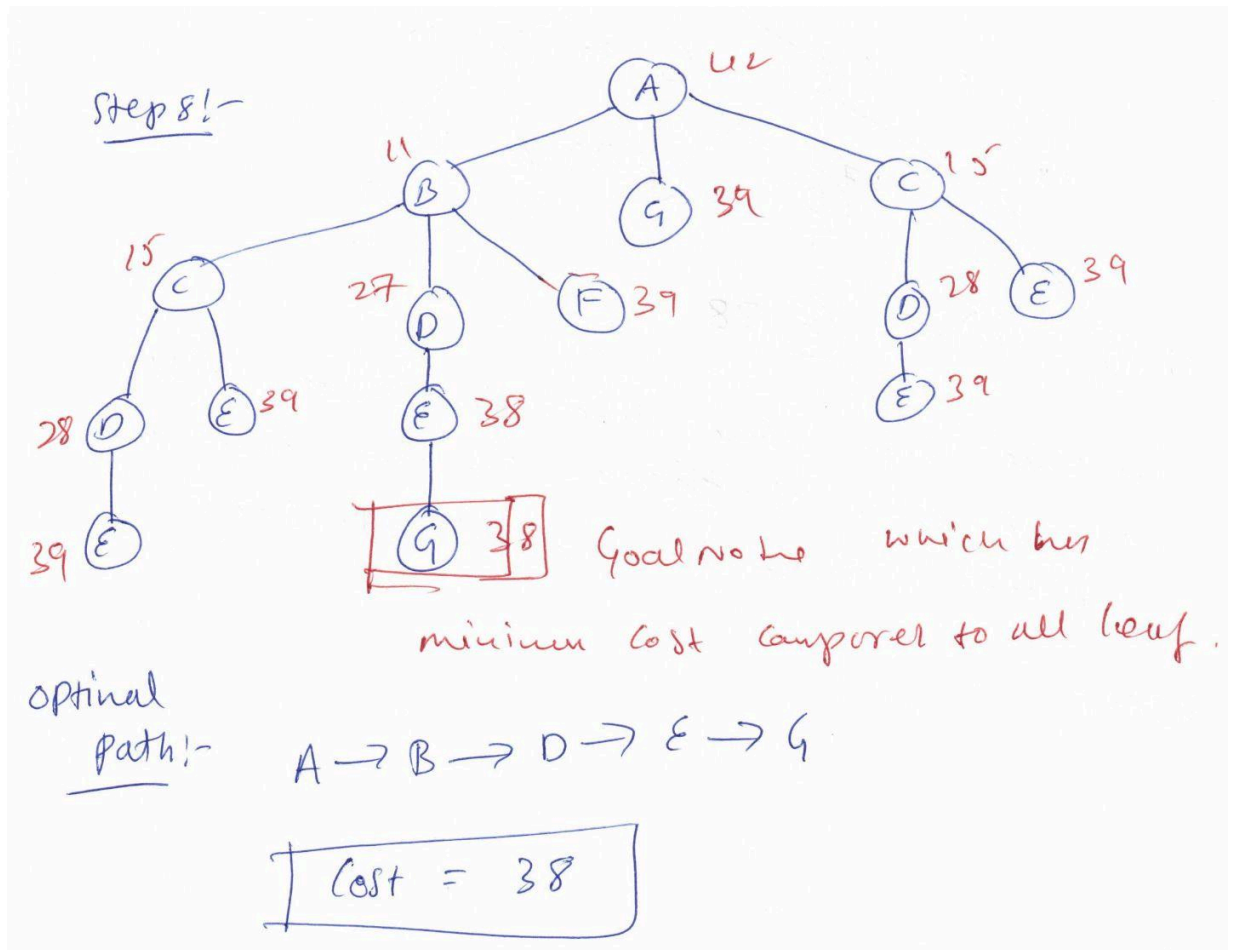
Step 3:- Expand 'C'



Step 4:- Expand 'C' which is child of 'A'







(Answers are marked in red colour)

3. (a) Satellite Image Analysis System is Fully Observable [True/False]
- (b) The action of the Simple reflex agent completely depends upon **Current percept**
- (c) What kind of environment is a crossword puzzle?
 - i) **Static** ii) Dynamic iii) Semi Dynamic iv) None of the mentioned
- (d) What is the heuristic function of greedy best-first search?
 - i) $f(n) \neq h(n)$, ii) $f(n) < h(n)$, **iii) $f(n) = h(n)$** , iv) $f(n) > h(n)$
- (e) Out of the given options, which of the following algorithms uses the least memory?
 - i) DFS** ii) BFS iii) Both DFS and BFS iv) None

(5 Marks)

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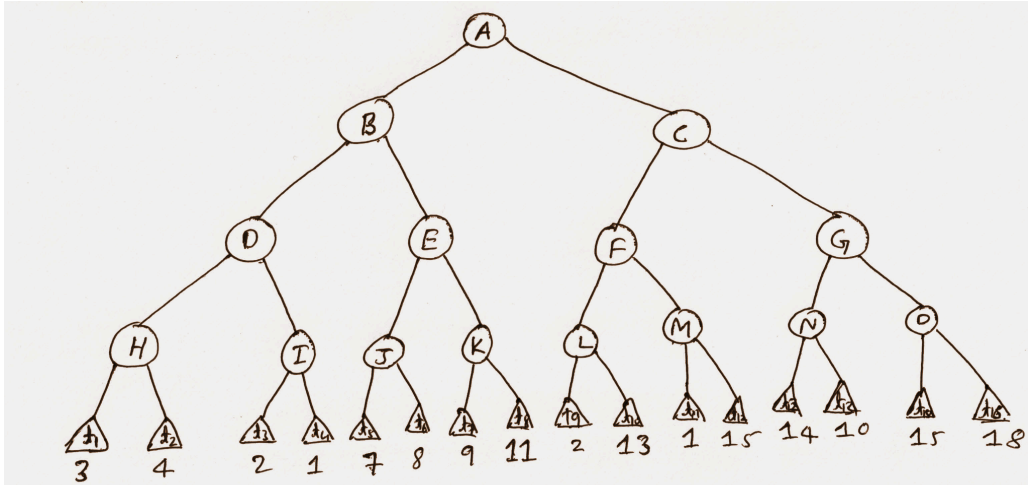
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY

TERM-II EXAMINATION – SPRING 2024**Artificial Intelligence****CSE: UG4 (PC)****Date: 14-03-2024****Duration: 90 Mins (03:30-05:00 PM)****Max. Marks: 25****Instructions:****Roll No:** _____

1. This is a **closed-book exam**. You can use a **calculator** if necessary.
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Section-A

1	a. Wumpus world is an episodic environment [True / False]. False b. Wumpus world is not a fully observable environment [True / False]. True	[2 Marks]
	c. Which of the following equivalences is true? (a) $1(P \wedge Q) \Leftrightarrow 1P \vee 1Q$ (b) $1(P \vee Q) \Leftrightarrow 1P \wedge 1Q$ (c) $1(P \rightarrow Q) \Leftrightarrow P \wedge 1Q$ (d) all of the above. (d) All the above	[3 Marks]
	d. The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below? I. $p \Rightarrow q$ II. $q \Rightarrow p$ III. $(\neg q) \vee p$ IV. $(\neg p) \vee q$ II and III	[2 Marks]
2	For a two-player game, a decision tree with utility values is given below. Find the best move for each player using the alpha-beta pruning technique. Note: Write each step in the text mentioning the order of traversal with node name, determination of alpha-beta values based on the decision, and pruned node name due to the algorithm. You may draw a single tree to show the alpha-beta values a	[8 Marks]

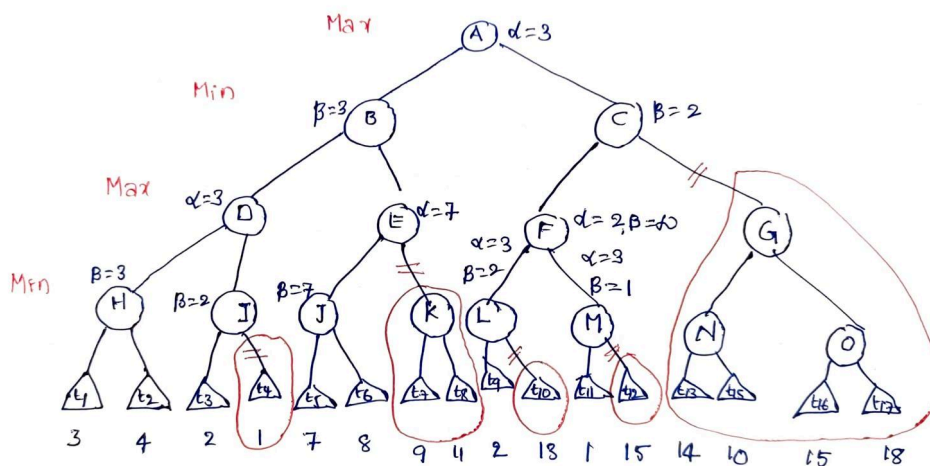


$\Delta \rightarrow$ terminal nodes (from left to right $t_1 - t_{16}$)

and the

pruned nodes.

However, an explanation for each node is mandatory. Marks will be awarded based on the explanation and justification, the tree diagram will be used for cross reference only.



3 Solve the following Cryptarithmic Problem. Show each step.

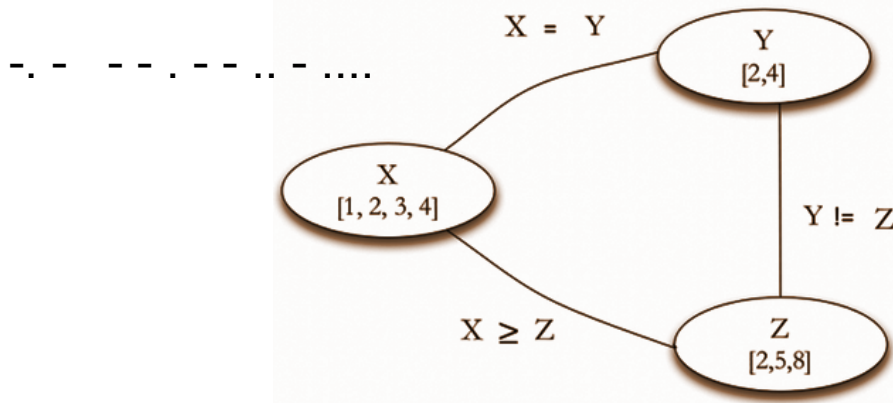
TWO
+TWO
FOUR

7 solutions.

ORWUTF = 483671
TWO 734
+ TWO 734
FOUR 1468

[4
M
ar
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	$\begin{array}{r} \text{ORWUTF} = 506371 \\ \text{TWO} \quad 765 \\ + \text{TWO} \quad 765 \\ \hline \text{FOUR} \quad 1530 \end{array}$ $\begin{array}{r} \text{ORWUTF} = 623781 \\ \text{TWO} \quad 836 \\ + \text{TWO} \quad 836 \\ \hline \text{FOUR} \quad 1672 \end{array}$ $\begin{array}{r} \text{ORWUTF} = 624981 \\ \text{TWO} \quad 846 \\ + \text{TWO} \quad 846 \\ \hline \text{FOUR} \quad 1692 \end{array}$ $\begin{array}{r} \text{ORWUTF} = 746381 \\ \text{TWO} \quad 867 \\ + \text{TWO} \quad 867 \\ \hline \text{FOUR} \quad 1734 \end{array}$ $\begin{array}{r} \text{ORWUTF} = 862591 \\ \text{TWO} \quad 928 \\ + \text{TWO} \quad 928 \\ \hline \text{FOUR} \quad 1856 \end{array}$ $\begin{array}{r} \text{ORWUTF} = 863791 \\ \text{TWO} \quad 938 \\ + \text{TWO} \quad 938 \\ \hline \text{FOUR} \quad 1876 \end{array}$ Any of the above answers will get 4 Marks.	
4	Solve the following constraint graph using AC-3 Algorithm.	[6 Marks]



Variables: {X, Y, Z}

Domain values are given inside the node. for e.g. $X = \{1, 2, 3, 4\}$

Constraints are given at the edges of the graph. For e.g. $\{Y \neq Z\}$

Note:

- In the First step, you have to specify all the variables and constraints and write all the arcs in the initial state of the queue in alphabetical order (like dictionary-based) only. (1M)*
Indicative example: $Queue = \{(A \neq B); (A \neq D); (A > E); (B = A); \dots\}$
- You must specify all the arcs in the queue w.r.t. previous steps, queue status, and domain sets for each variable at each step. (5 M)*
- Marks will be appropriately deducted for not following the above instructions.*
- No partial marks will be given for the remaining steps if note (a) is not followed in the first step.*
- No partial marks will be given for the problem if the queue and domain set are not written for each step.*

Answer:

Evaluation Instructions:

- In the First step, students have to clearly specify all the variables and constraints and write all the arcs in the initial state of the queue in alphabetical order only. (1 Mark)*
- You must clearly specify all the arcs in the queue w.r.t. previous steps, queue status and domain sets for each variable at each step. (Each Step 0.25 Marks)*
- No partial marks will be given for the remaining steps if step 1 is not correct.*
- Marks will be appropriately deducted for incorrect steps.*

Key Answer:

Step 1: (1 Mark)

Variables:	{X, Y, Z}
Constraints:	$X=Y$; $X \geq Z$; $Y \neq Z$;
Arcs:	$X=Y$; $X \geq Z$; $Y=X$; $Y \neq Z$; $Z \leq X$; $Z \neq Y$

Queue (in Ascending order):

$X=Y$
 $X \geq Z$
 $Y=X$
 $Y \neq Z$
 $Z \leq X$
 $Z \neq Y$

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Variables	Domain Values
X	{1,2,3,4}
Y	{2,4}
Z	{2,5,8}

Step 2: Removed ($X=Y$) (0.5 Marks)

Queue (in Ascending order):

~~$X=Y$~~
 $X \geq Z$
 $Y=X$
 $Y \neq Z$
 $Z \leq X$
 $Z \neq Y$

Variables	Domain Values
X	{1,2,3,4}
Y	{2,4}
Z	{2,5,8}

Change in domain value of X= {2, 4}

Value 1 and 3 deleted

The queue is updated with all arcs (except with Y) in which X is on RHS $\{(Z \leq X)\}$

Step 3: Removed ($X \geq Z$) (0.5 Marks)

Queue (in Ascending order):

~~$X=Y$~~
 ~~$X \geq Z$~~
 $Y=X$
 $Y \neq Z$
 $Z \leq X$
 $Z \neq Y$
 $Z \leq X$

.....

Variables	Domain Values
X	{2,4}
Y	{2,4}
Z	{2,5,8}

Comments: No Change in the domain value of X

Step 4: Removed ($Y=X$) (0.5 Marks)

Queue (in Ascending order):

~~$X=Y$~~
 ~~$X \geq Z$~~
 ~~$Y=X$~~
 $Y \neq Z$
 $Z \leq X$
 $Z \neq Y$
 $Z \leq X$

Variables	Domain Values
X	{2,4}
Y	{2,4}
Z	{2,5,8}

Comments: No Change in the domain value of Y

Step 5: Removed ($Y \neq Z$) (0.5 Marks)

Queue (in Ascending order):

~~$X=Y$~~

~~$X \geq Z$~~
 ~~$Y = X$~~
 ~~$Y \neq Z$~~
 $Z \leq X$
 $Z \neq Y$
 $Z \leq X$

Variables	Domain Values
X	{2,4}
Y	{2,4}
Z	{2,5,8}

Comments: No Change

Step 6: Removed ($Z \leq X$) (0.5 Marks)

Queue (in Ascending order):

~~$X = Y$~~
 ~~$X \geq Z$~~
 ~~$Y = X$~~
 ~~$Y \neq Z$~~
 ~~$Z \leq X$~~
 $Z \neq Y$
 $Z \leq X$

Variables	Domain Values
X	{2,4}
Y	{2,4}
Z	{2,5,8}

Comments: Change in domain value of $Z = \{2\}$
 Value 5 & 8 are deleted

The queue is updated with all arcs (except with X) in which Z is on RHS $\{(Y \neq Z)\}$

Step 7: Removed ($Z \neq Y$) (0.5 Marks)

Queue (in Ascending order):

~~$X = Y$~~
 ~~$X \geq Z$~~
 ~~$Y = X$~~
 ~~$Y \neq Z$~~
 ~~$Z \leq X$~~

~~$Z \neq Y$~~
 ~~$Z \leq X$~~
 ~~$Y \neq Z$~~

Variables	Domain Values
X	{2,4}
Y	{2,4}
Z	{2}

Comments: No Change in domain value

Step 7: Removed ($Z \leq X$) (0.5 Marks)

Queue (in Ascending order):

~~$X = Y$~~
 ~~$X \geq Z$~~
 ~~$Y = X$~~
 ~~$Y \neq Z$~~
 ~~$Z \leq X$~~
 ~~$Z \neq Y$~~
 ~~$Z \leq X$~~
 ~~$Y \neq Z$~~

Variables	Domain Values
X	{2,4}
Y	{2,4}
Z	{2}

Comments: No Change in domain value

Step 8: Removed $Y \neq Z$ (0.5 Marks)

Queue (in Ascending order):

~~$X = Y$~~
 ~~$X \geq Z$~~
 ~~$Y = X$~~
 ~~$Y \neq Z$~~
 ~~$Z \leq X$~~
 ~~$Z \neq Y$~~
 ~~$Z \leq X$~~
 ~~$Y \neq Z$~~

Variables	Domain Values
X	{2,4}
Y	{2,4}
Z	{2}

Comments: Change in domain value of $Y=\{4\}$
Value 2 is deleted

The queue is updated with all arcs (except with Z) in which Y is on RHS $\{(X=Y)\}$

Step 9: Removed $X=Y$ (0.5 Marks)

Queue (in Ascending order):

~~$X=Y$~~
 ~~$X \geq Z$~~
 ~~$Y=X$~~
 ~~$Y \neq Z$~~
 ~~$Z \leq X$~~
 ~~$Z \neq Y$~~
 ~~$Y=X$~~
 ~~$Z \neq Y$~~
 ~~$X=Y$~~

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Variables	Domain Values
X	{2,4}
Y	{4}
Z	{2}

Comments: Change in domain value of $X=\{4\}$
Value 2 is deleted

The queue is updated with all arcs (except with Y) in which X is on RHS $\{(Z \leq X)\}$

Step 10: Removed $Z \leq X$ (0.5 Marks)

Queue (in Ascending order):

~~$X=Y$~~
 ~~$X \geq Z$~~
 ~~$Y=X$~~
 ~~$Y \neq Z$~~
 ~~$Z \leq X$~~
 ~~$Z \neq Y$~~
 ~~$Y=X$~~
 ~~$Z \neq Y$~~
 ~~$X=Y$~~

Z←X	
Variables	Domain Values
X	{4}
Y	{4}
Z	{2}
Comments: No Change in domain value The queue is empty hence the algorithm stops. Final Domain Values are	
X	{4}
Y	{4}
Z	{2}



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY

TERM-III EXAMINATION – SPRING 2024**Artificial Intelligence****CSE: UG2 (PC)****Date: 24-04-2024****Duration: 90 Mins (03:30-05:00 PM)****Max. Marks: 25****Instructions:**

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1. This is a **closed-book exam**. You can use a **calculator** if necessary.
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Section-A

1	(a) What is the difference between clustering and classification ?	[3 Marks]
	(b) What is Reinforcement Learning? Explain with an Example.	[3 Marks]
2	Answer the following true/false. Solve step by step. (a) $\forall xP(x) \Leftrightarrow \neg \exists x \neg P(x)$ (b) $\exists xP(x) \Leftrightarrow \neg \forall x \neg P(x)$	[4 Marks]
3	a) Write the given sentences (I to VI) in first-order logic. (3M) b). Convert the sentences written in first-order logic (question 3. a)) to the Conjunctive Normal Form. (3M) c). Apply resolution to prove that “Scrooge is not a child” and draw the resolution graph. (4M) i. Every child loves Santa. ii. Everyone who loves Santa loves any reindeer. iii. Rudolph is a reindeer, and Rudolph has a red nose. iv. Anything which has a red nose is weird or is a clown. v. No reindeer is a clown. vi. Scrooge does not love anything which is weird. Prove by resolution graph that: Scrooge is not a child. Note: This is a connected question if a) is correct then b) will be evaluated, and if a) and b) both are correct then c) will be evaluated.	[10 Marks]

4	Suppose an RT-PCR test for the SARS-COV2 (COVID-19)-XE variant virus is 95% accurate. The test gives a positive result for 95% of those taking the test who are COVID-19 positive. Also, the test gives a negative result for 95% of those taking the test who are not COVID-19 positive. In Country X, forty percent of the residents are COVID-19 positive. Suppose a random resident of Country X takes the RT-PCR test and tests positive, what is the probability using Bayes theorem that the person is infected given that they have tested positive?	[5 Marks]
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Key Answers

1. (a)

CLASSIFICATION	CLUSTERING
used for supervised learning	used for unsupervised learning
process of classifying the input instances based on their corresponding class labels	grouping the instances based on their similarity without the help of class labels
it has labels so there is need of training and testing dataset for verifying the model created	there is no need of training and testing dataset

1 (b)

Reinforcement learning is an autonomous, self-teaching system that essentially learns by trial and error. It performs actions with the aim of maximizing rewards, or in other words, it is learning by doing in order to achieve the best outcomes.

Example:

1. Robotics: Robots with pre-programmed behavior are useful in structured environments, such as the assembly line of an automobile manufacturing plant, where the task is repetitive in nature.
2. A master chess player makes a move. The choice is informed both by planning, anticipating possible replies and counter replies.
3. An autonomous driving system must perform multiple perception and planning tasks in an uncertain environment. Some specific tasks where RL finds applications include vehicle path planning and motion prediction.

2. a) True b) False

3. Conversion of each sentence into FOL will get 0.5 Mark

i. Every child loves Santa.

$\forall x (CHILD(x) \rightarrow LOVES(x, Santa))$

ii. Everyone who loves Santa loves any reindeer.

$\forall x (LOVES(x, Santa) \rightarrow \forall y (REINDEER(y) \rightarrow LOVES(x, y)))$

iii. Rudolph is a reindeer, and Rudolph has a red nose.

$REINDEER(Rudolph) \wedge REDNOSE(Rudolph)$

iv. Anything which has a red nose is weird or is a clown.

$\forall x (REDNOSE(x) \rightarrow WEIRD(x) \vee CLOWN(x))$

v. No reindeer is a clown.

$\neg \exists x (REINDEER(x) \wedge CLOWN(x))$

or

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$\forall x (REINDEER(x) \rightarrow \neg CLOWN(x))$

vi. Scrooge does not love anything which is weird.

$\forall x (WEIRD(x) \rightarrow \neg LOVES(Scrooge, x))$

b). Conversion of each sentence into CNF will get 0.5 Mark

i. $\forall x (CHILD(x) \rightarrow LOVES(x, Santa))$

$R1: \neg CHILD(x) \vee LOVES(x, Santa)$

ii. $\forall x (LOVES(x, Santa) \rightarrow \forall y (REINDEER(y) \rightarrow LOVES(x, y)))$

$R2: \neg LOVES(p, Santa) \vee \neg REINDEER(q) \vee LOVES(p, q)$

iii. $REINDEER(Rudolph) \wedge REDNOSE(Rudolph)$

$R3: REINDEER(Rudolph)$

$R4: REDNOSE(Rudolph)$

iv. $\forall x (REDNOSE(x) \rightarrow WEIRD(x) \vee CLOWN(x))$

$R5: \neg REDNOSE(s) \vee WEIRD(s) \vee CLOWN(s)$

v. $\neg \exists x (REINDEER(x) \wedge CLOWN(x))$

$R6: \neg REINDEER(t) \vee \neg CLOWN(t)$

vi. $\forall x (WEIRD(x) \rightarrow \neg LOVES(Scrooge, x))$

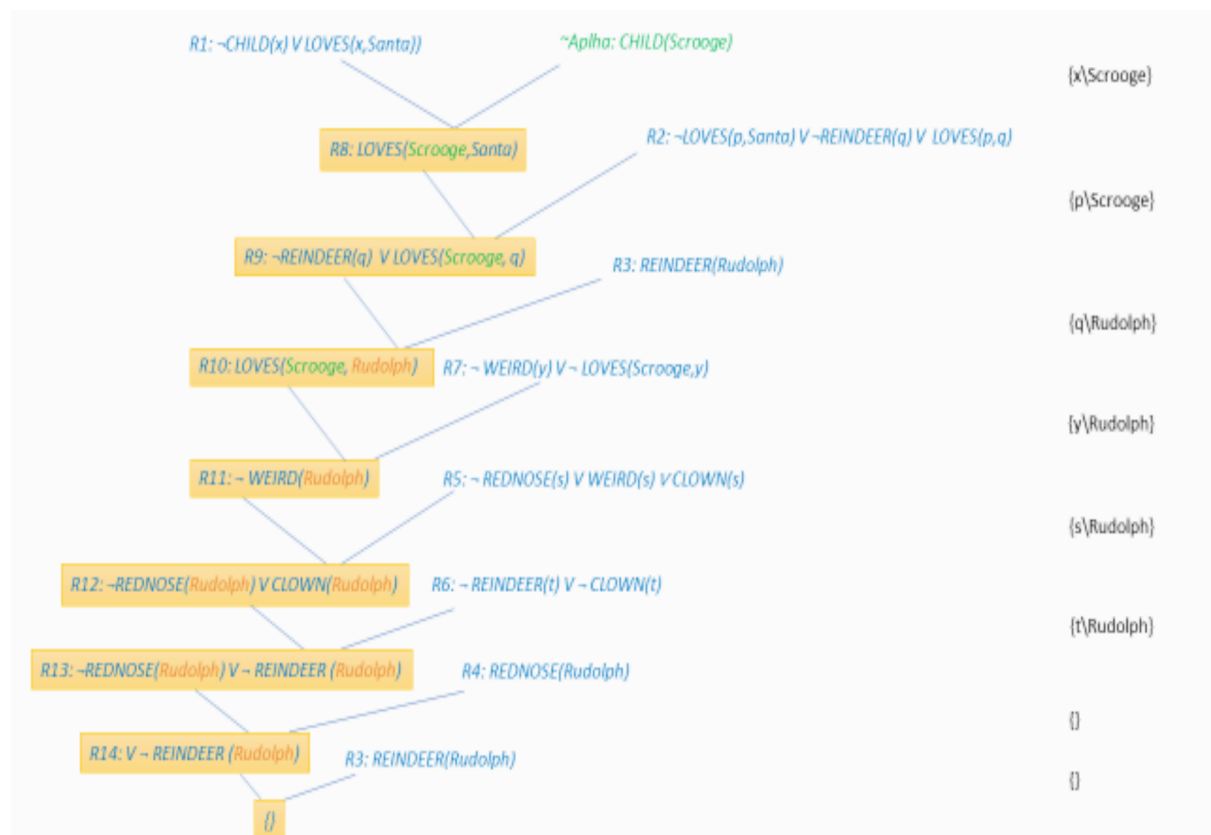
R7: $\neg WEIRD(y) \vee \neg LOVES(Scrooge, y)$

c) alpha: Scrooge is not a child. (Proof by a graph with at least 4 steps get full marks otherwise 1 mark for each resolvent step)

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$\neg CHILD(Scrooge)$

Negation of alpha ($\sim\alpha$) : $CHILD(Scrooge)$



The statement "Scrooge is not a child" is True.

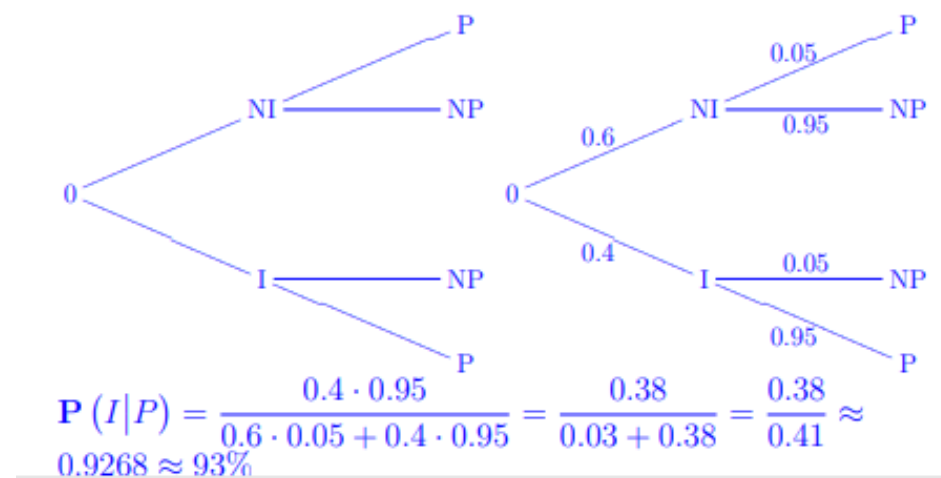
4. "O" total population

"I" Infected

"NI" Not Infected

"P" Positive

"NP" not positive



They need to calculate Infected given that Tested positive

I.e, $P(I|P)$

Note: If they have not used Baye's theorem and still calculated the answer then deduct 2.5 Marks for the correct answer.

If they solved correctly using Bayes' 5M

Incorrect answers get 0M